

TALLER: PROLOG

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#Ejemplo 1 clasificacion binaria
import numpy as np
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
```

```
y_true = np.array([3.0, -0.5, 2.0, 7.0, 4.5])
#y_pred = np.array([2.5, 0.0, 2.1, 7.2, 4.2])
#y_pred = np.array([2.9, 0.0, 1.9, 7.2, 4.1])
y_pred = np.array([3.0, -0.3, 2.0, 6.9, 4.5])
```

```
mse = mean_squared_error(y_true, y_pred)
print("Mean Squared Error:", mse)
rmse = np.sqrt(mse)
print("Root Mean Squared Error:", rmse)
mae = mean_absolute_error(y_true, y_pred)
print("Mean Absolute Error:", mae)
r2 = r2_score(y_true, y_pred)
print("R-squared:", r2)
```

```
Mean Squared Error: 0.009999999999999986
Root Mean Squared Error: 0.09999999999999994
Mean Absolute Error: 0.05999999999999993
R-squared: 0.9984025559105432
```

```
# EJEMPLO 2: REGRESIÓN (predicción de precios)
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
```

```
y_true_reg = np.array([3.0, -0.5, 2.0, 7.0, 4.5])
y_pred_reg = np.array([2.5, 0.0, 2.1, 7.2, 4.2])
```

```
mse = mean_squared_error(y_true_reg, y_pred_reg)
rmse = np.sqrt(mse)
mae = mean_absolute_error(y_true_reg, y_pred_reg)
r2 = r2_score(y_true_reg, y_pred_reg)
```

```
print("\n=== Métricas Regresión ===")
print(f"MSE : {mse:.4f}")
print(f"RMSE: {rmse:.4f}")
print(f"MAE : {mae:.4f}")
print(f"R² : {r2:.4f}")
```

```
=== Métricas Regresión ===
MSE : 0.1280
RMSE: 0.3578
MAE : 0.3200
R² : 0.9796
```

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